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CS-340: Client/Server Development
Professor Salem Othman
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Project Two: README

Project Overview

The purpose of this project is to develop a fully functional, interactive website that can display animal data in the form of a data table, geolocation chart, and a secondary chart for an international rescue animal training company called Grazioso Salvare. The website should allow its users to access a database that contains general information(i.e. breed, name, sex, etc.) for a number of dogs located among five different animal shelters within the region around Austin, Texas. Grazioso Salvare is relying on Global Rain(the software engineering company that I work for), to assist in gathering all the dogs that would be eligible for search-and-rescue training with the data provided to Global Rain through developing interactive options(i.e. Radio buttons, drop-down, etc.) to filter the data. In order for a dog to be eligible for search-and-rescue training, characteristics such as breed and age play a major role in their effectiveness. There are also breeds that are proficient in unique domains such as water rescue, mountain/wilderness rescue, and disaster rescue. It is my job to identify a list of dogs that meet the criteria specified by Grazioso Salvare.

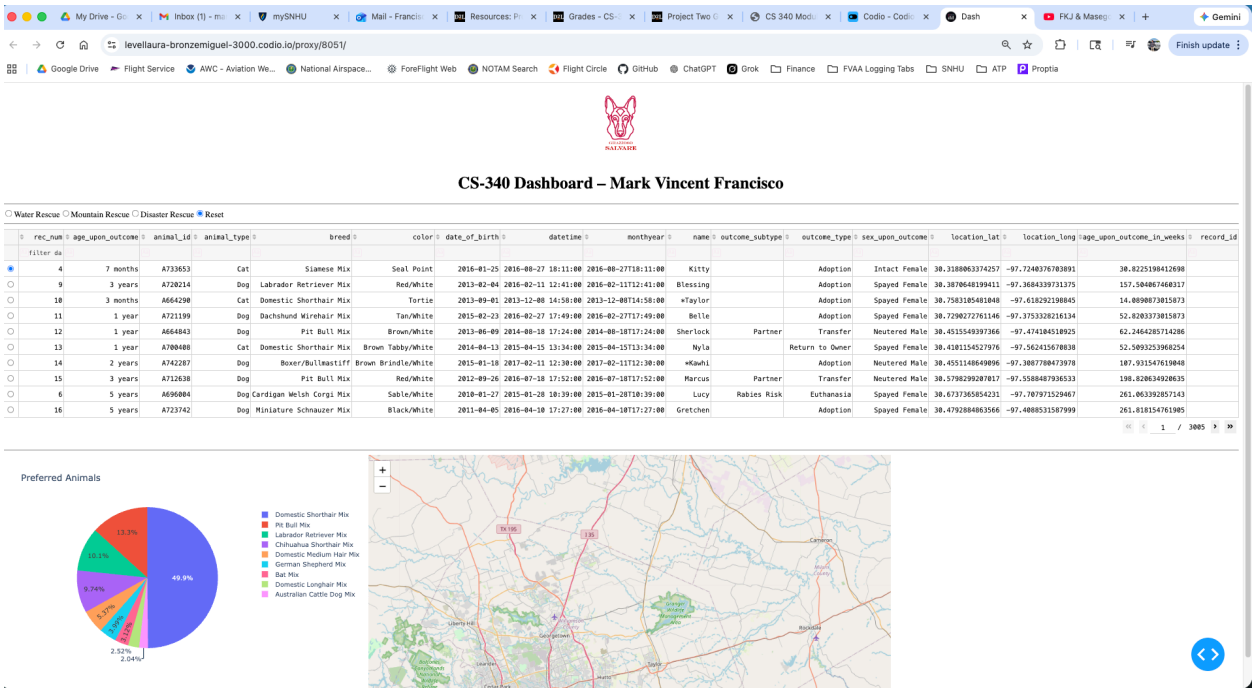
In order for the data to be displayed in the form of a data table, geolocation chart, and pie chart, it is important to incorporate CRUD(create, read, update, and delete) operations onto a MongoDB database by using the python programming language. These functions are within the CRUD python module that was developed in previous milestones of this course.

Required Functionality

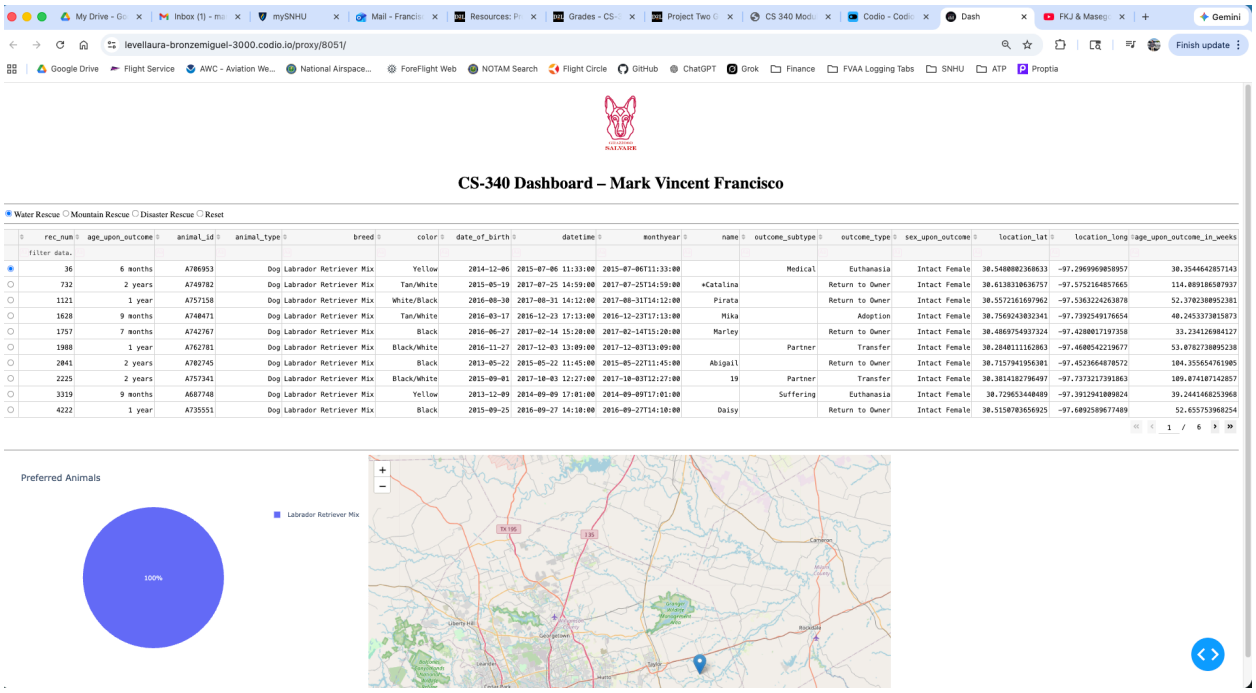
1. Branding Requirements
    a. Displaying the Grazioso Salvare logo that is linked to https://www.snhu.edu as a URL upon click.
    b. Implementing a unique author identifier by using my name, "CS-340 Dashboard – Mark Vincent Francisco".
2. Interactive Data Table
    a. Displays the shelters animals among the five animal shelters within the Austin, Texas area.
    b. Updates the table based on the selected filter.
3. Geolocation chart
    a. Displays the location of a selected animal on a map at the bottom of the website.
    b. Updates upon selecting a new row in the above data table.
4. Secondary Chart (Pie chart)
    a. Displays the top 9 breeds based on the filtering data.
    b. Updates upon changing the filter option as well.
5. Interactive Filtering Options
    a. Allowing the user to filter dogs from a number of radio buttons including:
        i. Water rescue
        ii. Mountain rescue
        iii. Disaster rescue
        iv. Reset
    b. The above filtering options are based on:
        i. Preferred Breeds
        ii. Preferred Sex
        iii. Training Age

Screenshots of Functionality

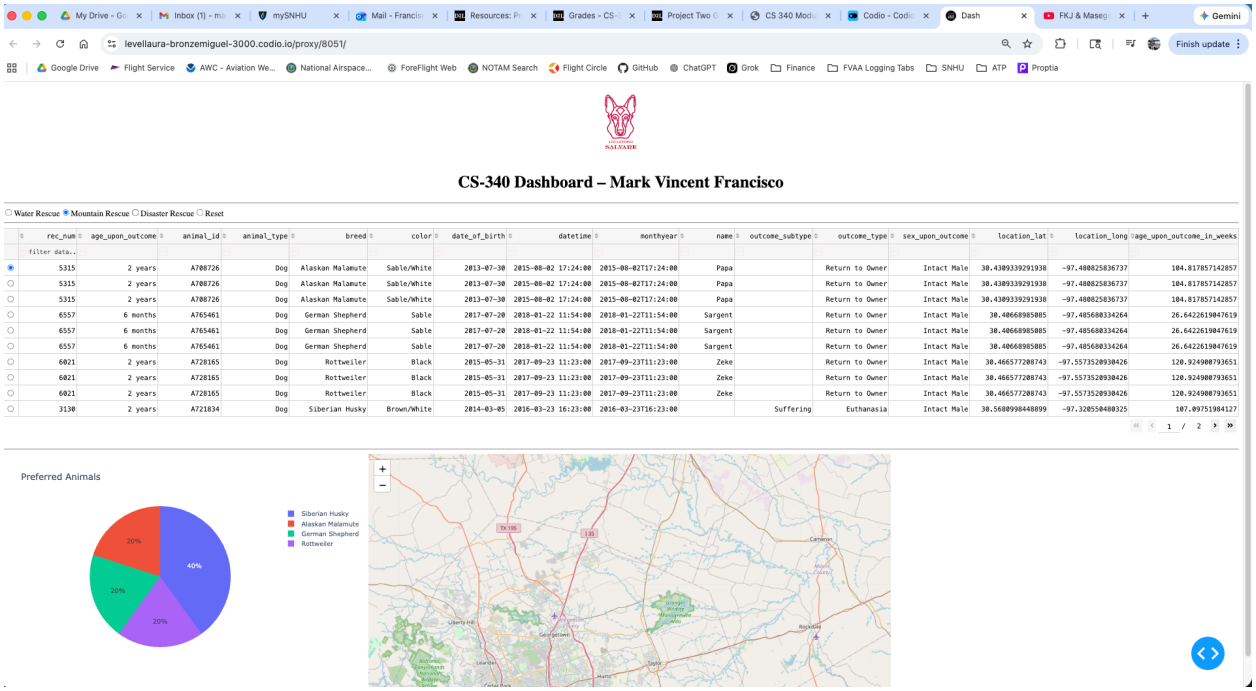
1. Reset Filter Applied



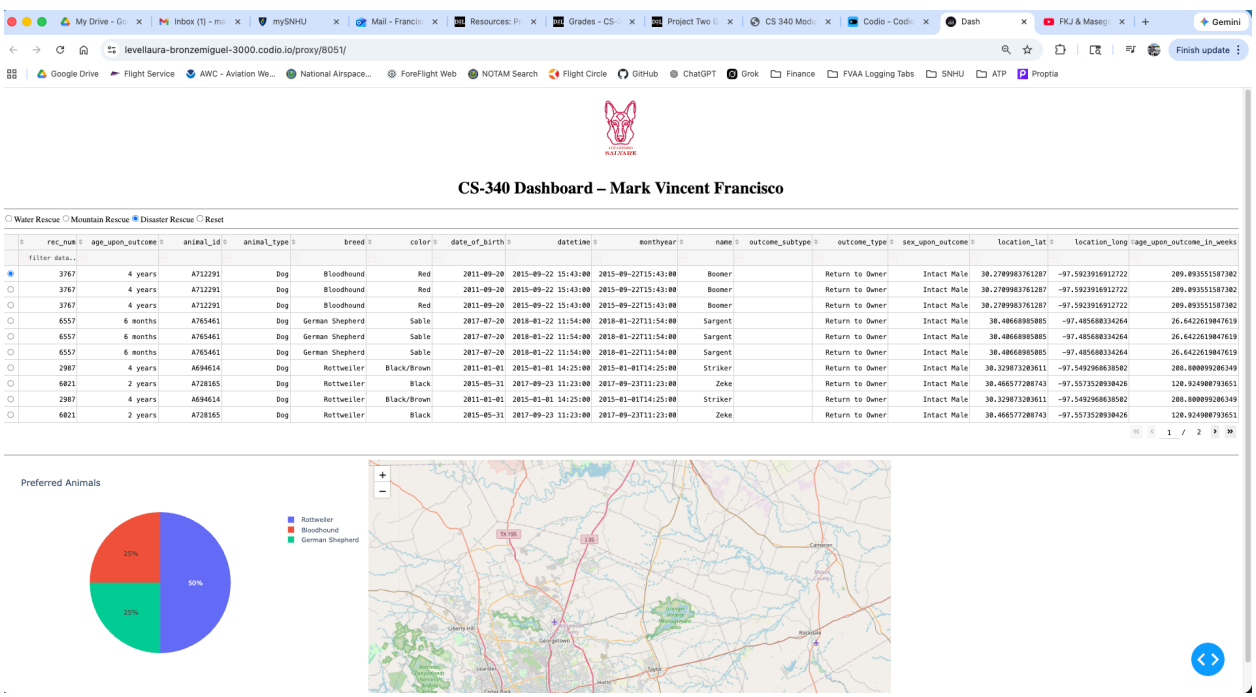
2. Water Rescue Filter Applied



3. Mountain Rescue Filter Applied



4. Disaster Rescue Filter Applied



Tools Used and Rationale

1. MongoDB
    a. Used as a model component of development because:
        i. It is an excellent framework for high flexibility, scalability and performance.
        ii. It stores data as BSON documents for effective querying and indexing.
        iii. It integrates well with Python through the PyMongo driver. Using this driver integrates all CRUD functionality, as well as seamless interaction between the MongoDB and python dictionaries/lists.
2. Dash Framework
    a. Used for its view and controller structure for the web application because it provides:
        i. A pie chart for data visualization
        ii. A geolocation map
        iii. A dash table
        iv. Interactive Radio buttons for effective filtering
3. Python
    a. Primary programming language to construct the CRUD python module.
4. CRUD Module containing the AnimalShelter Class
    a. Supports the Create, Read, Update, and Delete operations.
    b. The controller for Dash and MongoDB.
5. Additional Libraries Use
    a. Base64 – for the Grazioso Salvare logo
    b. Pandas – for data handling
    c. Plotly Express – used for the pie chart
    d. Dash Leaflet – used for the geolocation map

Instructions to Reproduce the Project

1. First, create a MongoDB through the terminal by using the appropriate commands.
2. Create a python module file containing the PyMongo library and the class with all necessary CRUD functions for the MongoDB created.
3. Create a dashboard script file containing following widgets/features:
    a. Logo with a unique identifier(name, photo)
    b. Radio buttons
        i. Filter by age, breed and gender.
    c. Data Table
    d. Geolocation Chart
    e. Pie Chart
4. Test all filter options and check if there were any changes to the data table and charts.
5. Take screenshots of each filter option selected affecting the data table, pie chart, and geolocation map.

Challenges Encountered

1. Index errors occurring within the map callback
    a. Whenever selecting a filter option, an index out of range error appears at the top right of the site. The solution to address this error was by implementing row validation logic within the update_map function to prevent these errors from occurring.

2. Syntax issues when querying the MongoDB
    a. Initially used python sets instead of the MongoDB $in operator. I then replaced the sets with: {“breed”: {“$in”: [...]}}

Conclusion

Completing this project resulted in a user-friendly dashboard containing interactive filtering buttons, a mutable data table based on filter changes, and informative visuals such as the pie chart and geolocation map. The features contained within the dashboard integrated well with MongoDB, Python, and Dash within an MVC design structure. The completed dashboard allows Grazioso Salvare to quickly identify the dogs that are eligible for search-and-rescue training based on their required criteria using the data provided.